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June 26, 2026

The Editors
Scientific Reports

Dr Arumugam,

I am pleased to submit my manuscript, “A decentralized neutrosophic decision layer for global-state freshness in microservices architectures,” for consideration as an Article in *Scientific Reports*.

The problem. Microservices routinely make business decisions over global state—a price, a fraud flag, a model feature—while each service is sovereign over its own data and the only cross-service signal is a cache that may hold values that were never committed. The prevailing answer collapses this to a boolean fresh/stale flag, which cannot distinguish a value held only in cache (unconfirmed) from one that is genuinely stale or absent.

The contribution. The manuscript develops a decentralized decision layer in which each data item carries a single-valued neutrosophic triple (T, I, F) for its persistence status; peers fuse their views with weighted operators and a deneutrosophy score, and then act or abstain with no central authority. To my knowledge this is the first work to turn this representation into a concrete, formally analysed, and reproducibly evaluated mechanism. I provide one-round termination and safety/liveness guarantees generalized to a graded encoding, a Byzantine analysis that separates gate robustness from value selection (including a proven limitation and the precise fix it motivates), and a controlled evaluation against strong, eventual, quorum, and single-probability baselines across four scenarios—including a distinct ML feature-store domain—on a reproducible simulator and a real, emulated-WAN container testbed.

Why *Scientific Reports*. The work is a cross-disciplinary systems contribution: it bridges neutrosophic decision theory, distributed-systems consistency, and reproducible empirical methodology, and it is written to be accessible to a broad readership. I emphasise transparent, calibrated reporting—every reported number regenerates end-to-end from committed code, a content-hashed calibration, and recorded seeds. I make no superiority claim over consensus protocols; the manuscript characterises a calibrated trade-off and reports null results (for example, that a calibrated single-probability gate is competitive) plainly.

Statements. This manuscript is original, is not under consideration elsewhere, and has not been discussed with a member of the journal’s editorial board prior to submission. The author declares no competing interests and received no specific funding for this work.

Thank you for considering my work.

Regards,

Haitham A. El-Ghareeb